

1 Solution — GIAM 7.1

Problem 4

1.1 Problem

Prove that the binomial coefficients

$$\binom{n+k-1}{k} \quad \text{and} \quad \binom{n+k-1}{n-1}$$

are equal.

1.2 Setup

Throughout, n and k are non-negative integers with $n \geq 1$ (so that $n-1 \geq 0$). Recall that for non-negative integers m, r with $0 \leq r \leq m$,

$$\binom{m}{r} = \frac{m!}{r!(m-r)!}.$$

Set $m := n + k - 1$, and observe the crucial arithmetic fact

$$k + (n - 1) = n + k - 1 = m,$$

so that $n - 1 = m - k$. The claim is therefore the *symmetry identity*

$$\binom{m}{r} = \binom{m}{m-r}$$

specialized to $m = n + k - 1$ and $r = k$. We give two proofs — one by direct computation, one by an explicit bijection.

1.3 Proof 1 — Algebraic

Apply the factorial formula to each side:

$$\binom{n+k-1}{k} = \frac{(n+k-1)!}{k!((n+k-1)-k)!} = \frac{(n+k-1)!}{k!(n-1)!},$$

$$\binom{n+k-1}{n-1} = \frac{(n+k-1)!}{(n-1)!((n+k-1)-(n-1))!} = \frac{(n+k-1)!}{(n-1)!k!}.$$

The two right-hand sides differ only by the order of the factors in the denominator. Since multiplication of natural numbers is commutative, $k! \cdot (n-1)! = (n-1)! \cdot k!$, so

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}. \quad \blacksquare$$

1.4 Proof 2 — Combinatorial Bijection

Let S be any set of size $m = n + k - 1$. By definition,

$$\binom{m}{k} = \#\{A \subseteq S : |A| = k\}, \quad \binom{m}{n-1} = \#\{B \subseteq S : |B| = n-1\}.$$

Define the *complement* map

$$\varphi : \{k\text{-subsets of } S\} \longrightarrow \{(n-1)\text{-subsets of } S\}, \quad \varphi(A) = S \setminus A.$$

Two checks:

1. **Well-defined.** For any k -subset $A \subseteq S$, the complement $S \setminus A$ has size $|S| - |A| = m - k = (n + k - 1) - k = n - 1$, so $\varphi(A)$ really is an $(n-1)$ -subset of S .
2. **Bijective.** Applying φ twice yields $\varphi(\varphi(A)) = S \setminus (S \setminus A) = A$. Hence φ is its own inverse, and therefore a bijection.

A bijection between two finite sets equates their cardinalities, so

$$\binom{n+k-1}{k} = \#\{k\text{-subsets of } S\} = \#\{(n-1)\text{-subsets of } S\} = \binom{n+k-1}{n-1}. \quad \blacksquare$$

1.5 Remark — Stars and Bars

The two expressions $\binom{n+k-1}{k}$ and $\binom{n+k-1}{n-1}$ are the two standard forms of the *stars-and-bars* coefficient, which counts the number of multisets of size k drawn from an n -element set. To see the connection, arrange k identical “stars” together with $n-1$ identical

“bars” in a row of total length $n + k - 1$; the bars partition the stars into n groups, one per underlying element, and the number of stars in each group records that element’s multiplicity. The total number of such arrangements can be counted in two equivalent ways:

- **Place the stars.** Choose which k of the $n + k - 1$ positions are stars: $\binom{n+k-1}{k}$ arrangements.
- **Place the bars.** Choose which $n - 1$ of the $n + k - 1$ positions are bars: $\binom{n+k-1}{n-1}$ arrangements.

These two counts agree precisely because of the identity proved above. Their common value is the *multiset coefficient*, sometimes written

$$\left(\binom{n}{k}\right) = \binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

1.6 Numerical Sanity Check

For $n = 4$, $k = 3$ — multisets of size 3 from $\{1, 2, 3, 4\}$ — both expressions should evaluate to the same number:

$$\binom{4+3-1}{3} = \binom{6}{3} = 20, \quad \binom{4+3-1}{4-1} = \binom{6}{3} = 20.$$

For $n = 5$, $k = 2$ — multisets of size 2 from $\{1, \dots, 5\}$:

$$\binom{5+2-1}{2} = \binom{6}{2} = 15, \quad \binom{5+2-1}{5-1} = \binom{6}{4} = 15.$$

Both check out — and the value 15 matches the direct count of size-2 multisets from a 5-set: the $\binom{5}{2} = 10$ unordered pairs of distinct elements plus the 5 “double” multisets $\{x, x\}$, totalling 15.